

Price dynamics via the Langevin-Helmholtz equation

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The model proposes that the price dynamics of financial assets are not a pure random walk but a succession of latent (primitive) states that manifest themselves noisily. Assets are classified by the "signature" of their confinement potentials, where liquidity acts as the damping/diffusion parameter in the system.

Introduction

Financial prices emerge from interacting heterogeneous agents and appear stochastic at the aggregate level. Their apparent randomness, together with persistent regularities, may invite analogy with thermodynamic variables governed by invariant laws. Yet unlike temperature or pressure, prices arise from adaptive agents whose learning and strategic interaction introduce reflexivity and structural instability; observed regularities therefore reflect evolving collective organization rather than fixed microscopic laws.

This tension has shaped quantitative finance. Bachelier's Brownian diffusion [1] formalized Gaussian independent increments, whereas Mandelbrot [2] demonstrated heavy tails and scaling inconsistent with linear diffusion. Modern factor theory shows that empirical covariance matrices admit approximate low-rank structure, theoretically grounded by Chamberlain and Rothschild (1983) [3], empirically supported by Fama and French (1993) [4], and spectrally confirmed through random matrix analysis (Laloux et al., 1999 [5]; Plerou et al., 1999 [6]). Financial systems thus organize around dominant collective modes.

We propose that observable prices are projections of a stochastic process evolving in a latent space spanned by structural *primitives*, defined as eigenfunctions of an integral operator K [7],

$$K\psi_i = \lambda_i\psi_i. \quad (1)$$

Volatility clustering and regime shifts correspond to persistence and spectral reconfiguration of K . Diffusion, scaling laws, and factor structures emerge as reduced-form manifestations of an underlying operator-driven geometry.

The question is therefore not whether markets are random, but in which spectral structure that randomness unfolds.

Formulation

Consider a finite time window $t \in [0, T]$ and define log-returns

$$P(t) = \ln \left(\frac{S(t)}{S(t - \Delta t)} \right), \quad (2)$$

assumed centered within each window. For a given regime r , define the temporal covariance kernel

$$K^{(r)}(t, t') = \langle P(t)P(t') \rangle^{(r)}, \quad (3)$$

which induces a compact, self-adjoint positive operator on $L^2([0, T])$,

$$(\mathcal{C}^{(r)}f)(t) = \int_0^T K^{(r)}(t, t')f(t')dt'. \quad (4)$$

The structural modes are obtained from the spectral problem

$$\int_0^T K^{(r)}(t, t')\phi_k^{(r)}(t')dt' = \lambda_k^{(r)}\phi_k^{(r)}(t). \quad (5)$$

This decomposition is the functional form of Principal Component Analysis (PCA): the eigenfunctions extremize projected variance and provide the optimal rank- m approximation in mean-square norm. In discrete time, this construction reduces to the spectral decomposition of the empirical covariance matrix $\hat{C} = (1/N)X^\top X$, with X being the matrix of log-returns.

The kernel formulation used here is the continuum limit of that representation, with eigenfunctions acting as time-resolved principal components. Within the window, returns admit the expansion

$$P(t) = \sum_{k=1}^m a_k^{(r)}\phi_k^{(r)}(t) + \eta^{(r)}(t), \quad (6)$$

where the amplitudes $a_k^{(r)}$ are fixed with respect to the internal time t of the window. In high dimension, the empirical operator fluctuates around its population counterpart. Even in the absence of genuine structure, $\hat{C}$ exhibits a continuous eigenvalue bulk governed by random matrix theory. Structural modes correspond to Eigenvalues that detach from this bulk, while modes within it

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represent universal finite-sample fluctuations. The separation between informative and noise subspaces is therefore intrinsically spectral and dimension-dependent. If the regime persists, the operator $K^{(r)}$ and its spectrum remain unchanged and only the amplitudes update. If $r \rightarrow r'$, the kernel transforms

$$K^{(r)} \longrightarrow K^{(r')}, \quad (7)$$

and its spectral geometry is modified. In this framework, a market regime is identified with the spectral structure of the covariance operator, and regime transitions correspond to spectral reconfiguration rather than discrete Markov switching.

Spectral Dynamics of the Covariance Operator
Let the covariance kernel depend on a slow window index τ ,

$$K(t, t'; \tau),$$

where $t \in [0, T]$ denotes the internal time and τ indexes successive windows. The eigenfunctions satisfy

$$\int_0^T K(t, t'; \tau) \phi_k(t'; \tau) dt' = \lambda_k(\tau) \phi_k(t; \tau). \quad (8)$$

If the regime evolves slowly, K is differentiable in τ . Standard perturbation theory for self-adjoint operators yields the adiabatic evolution

$$\frac{\partial \phi_k}{\partial \tau} = \sum_{j \neq k} \frac{\langle \phi_j, (\partial_\tau K) \phi_k \rangle}{\lambda_k - \lambda_j} \phi_j, \quad (9)$$

analogous to adiabatic spectral rotation. When eigenvalues approach degeneracy ($\lambda_k \approx \lambda_j$), the basis rotates strongly, signaling structural instability and regime transition.

If instead the kernel undergoes a jump at τ_0 ,

$$K(\tau_0^+) = K(\tau_0^-) + \Delta K, \quad (10)$$

the spectral basis rotates instantaneously via an orthogonal transformation, corresponding to a non-adiabatic regime shift.

We model the full regime dynamics as an Itô process with jumps in the space of kernels,

$$dK = A[K] d\tau + B[K] dW_\tau + \Delta K dN_\tau, \quad (11)$$

where the drift-diffusion term generates slow spectral rotation within a regime, and the Poisson jump term captures abrupt structural transitions.

Dynamics of the Mode Amplitudes

Within a window, returns admit the expansion

$$P(t) = \sum_k a_k(\tau) \phi_k(t; \tau),$$

with

$$a_k(\tau) = \int_0^T P(t) \phi_k(t; \tau) dt. \quad (12)$$

As the basis rotates with the slow time τ , the amplitudes evolve according to

$$\frac{da_k}{d\tau} = \sum_{j \neq k} \Omega_{kj}(\tau) a_j(\tau), \quad (13)$$

where

$$\Omega_{kj}(\tau) = \frac{\langle \phi_j, (\partial_\tau K) \phi_k \rangle}{\lambda_k - \lambda_j}.$$

Thus, the coefficients obey a coupled linear system: slow spectral rotation of K induces energy transfer between modes. As eigenvalues approach degeneracy, $|\lambda_k - \lambda_j| \rightarrow 0$, the coupling strengthens, leading to mode mixing, growth of the residual component, and loss of PCA stability preceding regime transition.

If instead the kernel undergoes a jump,

$$K^+ = K^- + \Delta K,$$

the eigenbasis rotates instantaneously,

$$\phi_k^+ = \sum_j M_{kj} \phi_j^-,$$

and the amplitudes transform accordingly,

$$a_k^+ = \sum_j M_{kj} a_j^-, \quad (14)$$

corresponding to a non-adiabatic redistribution of modal energy.

Noise, Random Matrix Bulk, and Spectral Transitions

The residual term η is not arbitrary: it corresponds to the universal random matrix noise predicted for finite-sample covariance operators. The full spectral decomposition of the kernel reads

$$K(t, t') = \sum_{k=1}^{\infty} \lambda_k \phi_k(t) \phi_k(t').$$

We separate structural modes ($\lambda_k > \lambda^+$) from bulk modes ($\lambda_k \leq \lambda^+$), where the Marchenko-Pastur upper edge is

$$\lambda^+ = \sigma^2(1 + \sqrt{q})^2,$$

with $q = L/N$ the window aspect ratio [5, 6]. Empirically, eigenvalues inside this interval follow random matrix predictions [8].

Truncating to m structural modes,

$$P(t) = \sum_{k=1}^m a_k \phi_k(t) + \eta(t), \quad \eta(t) = \sum_{k>m} a_k \phi_k(t),$$

and since the basis diagonalizes the covariance,

$$\langle a_k a_j \rangle = \lambda_k \delta_{kj}.$$

For $k > m$, λ_k belongs to the MP bulk and

$$a_k \sim \mathcal{N}(0, \lambda_k),$$

independent to leading order. The residual covariance is therefore

$$\langle \eta(t) \eta(t') \rangle = \sum_{k>m} \lambda_k \phi_k(t) \phi_k(t').$$

As bulk eigenvectors are asymptotically random and uncorrelated [9, 10], the residual converges to

$$\langle \eta(t) \eta(t') \rangle \approx \sigma_\eta^2 \delta(t - t'),$$

i.e., temporally white noise within the window, with total variance

$$\sigma_\eta^2 = \sum_{k>m} \lambda_k.$$

Noise increases when structural eigenvalues approach the MP threshold, signaling loss of informative structure.

Regime Transitions as Spectral Phase Transitions

Within a regime, well-separated eigenvalues imply stable modes, since

$$\frac{\partial \phi_k}{\partial \tau} \propto \frac{1}{\lambda_k - \lambda_j}.$$

As $\lambda_k \rightarrow \lambda_j$, spectral degeneracy induces strong mode mixing, instability of the PCA basis, and growth of the residual.

We interpret regime transitions as spectral phase transitions of the covariance operator. Defining the spectral gap

$$\Delta(\tau) = \min_{k \neq j} |\lambda_k - \lambda_j|,$$

a transition occurs when $\Delta \rightarrow 0$, leading to reorganization of the dominant subspace. Additionally, eigenvalues may cross the MP threshold, altering the number of structural modes.

Thus, regime changes arise either from violent rotation of the dominant eigenspace or from structural modes entering or leaving the random matrix bulk, corresponding to variation in effective market information.

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